

Emanuele Rossi

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Researcher working on machine learning for animal communication, part of a broader interest in leveraging ML to recover structure in complex systems: previously social graphs (Temporal Graph Networks, SIGN) at Twitter and Imperial College London, and generative models for biomolecules at VantAI.

Education

Imperial College London <i>Ph.D. in Computer Science – Supervisor: Prof. Michael Bronstein</i>	<i>Oct 2020 – Feb 2024</i>
◦ Research focused on graph machine learning methods for large real-world systems	
University of Cambridge <i>Master in Advanced Computer Science, Distinction (4.0 GPA)</i>	<i>Oct 2018 – Jun 2019</i>
Imperial College London <i>Bachelor in Computer Science, First Class Honors (4.0 GPA)</i>	<i>Oct 2015 – Jun 2018</i>

Experience

Postdoctoral Researcher <i>GLADIA Research Group ↗, Sapienza University of Rome</i>	<i>Rome (remote)</i> <i>Oct 2025 – Present</i>
◦ Developing machine learning methods to decode animal communication. Supervisor: Prof. Emanuele Rodolà	
Machine Learning Researcher <i>VantAI</i>	<i>NYC (remote)</i> <i>Jan 2024 – Aug 2025</i>
◦ Co-led development of Neo-1 ↗ , a state-of-the-art all-atom latent diffusion model for multimodal structure prediction and de-novo generation of biomolecular complexes, trained on 100+ GPUs.	
Machine Learning Researcher <i>Twitter</i>	<i>London</i> <i>Jun 2019 – Feb 2023</i>
◦ Developed graph ML models for Twitter's social graph and deployed a recommendation model that generated 1.2M additional daily follows	
Machine Learning Researcher <i>Fabula AI</i>	<i>London</i> <i>Mar 2019 – Jun 2019</i>
◦ Developed graph ML models for fake news detection as part of Fabula AI's small research team before Twitter acquired the company ↗	
Software Engineering Intern <i>Google</i>	<i>California</i> <i>Jul 2017 – Oct 2017</i>

Selected Publications

[Full list](#)

Animal Communication & Audio

- E. Rossi and E. Rodolà. [Communicating Sound Through Natural Language \[↗\]\(#\)](#). *arXiv* 2026.
- D. Marincione et al. [Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models \[↗\]\(#\)](#). *NeurIPS Workshop on AI for Animal Communication 2025, Spotlight (top 15%)*.

Graph Learning

- E. Rossi et al. [Temporal Graph Networks for Deep Learning on Dynamic Graphs \[↗\]\(#\)](#). *ICML GRL Workshop 2020*. **1,700+ citations** and **1,000+ GitHub stars [↗](#)**.
- E. Rossi*, F. Frasca* et al. [SIGN: Scalable Inception Graph Neural Networks \[↗\]\(#\)](#). *ICML GRL Workshop 2020*. **600+ citations** and used in **production at Airbnb [↗](#)**.
- B. Chamberlain, S. Shirobokov, E. Rossi et al. [Graph Neural Networks for Link Prediction with Subgraph Sketching \[↗\]\(#\)](#). *ICLR 2023, Oral (top 5%)*. **200+ citations**.

Leadership

Co-Founder, LeadTheFuture (Forbes article ↗)	<i>Jun 2018 – Mar 2023</i>
◦ Built a non-profit mentoring community with more than 200 mentors and 500 selected mentees	